



Analysis of EEG Signals using Discrete Fourier Transform

Identification of Prominent Frequency Components and Signal Reconstruction

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1 Introduction

EEG signals reflect brain activity and consist of various frequency components linked to mental states. To analyze these, we apply the Discrete Fourier Transform (DFT), which converts the signal to the frequency domain.

In this work, we will:

1. Perform manual DFT on an EEG segment,
2. Identify the most prominent frequency bands,
3. Reconstruct the signal using these bands via manual IDFT,
4. Compare the reconstructed signal with the original.

This shows how basic frequency analysis helps understand EEG signals.

2 Motivation

Analyzing EEG in the frequency domain helps:

- Detect brain states like focus or relaxation,
- Support clinical diagnosis using brainwave patterns,
- Learn core signal processing concepts through manual coding.

This task offers practical insight into EEG analysis without relying on built-in functions.

3 Background Study

EEG signal analysis has long been used to study brain activity through characteristic frequency bands. Fourier analysis enables conversion of EEG signals from time to frequency domain for better interpretation.

Key EEG bands include:

- **Delta (0.5–4 Hz):** Deep sleep.
- **Theta (4–8 Hz):** Drowsiness, memory, meditation.
- **Alpha (8–13 Hz):** Relaxed wakefulness.
- **Beta (13–30 Hz):** Active thinking, focus.
- **Gamma (30–100+ Hz):** High-level cognitive functions.

This assignment focuses on analyzing an EEG segment using DFT to explore these bands and reconstruct the signal using the most dominant components.

4 Social and Economic Impact

EEG signal analysis has wide-reaching social and economic importance:

- **Healthcare:** Helps detect neurological disorders early, improving treatment outcomes and reducing costs.
- **Assistive Tech:** Enables Brain-Computer Interfaces (BCIs) that assist people with disabilities.
- **Mental Health:** Used in non-invasive therapies for stress, anxiety, and ADHD.
- **Productivity:** Better brain monitoring supports well-being and cognitive performance.

This work demonstrates basic tools that support these real-world applications.

5 Related Mathematical Studies

The analysis of EEG signals using DFT is grounded in several mathematical concepts from signal processing and Fourier analysis.

- **Discrete Fourier Transform (DFT):** The DFT is the version of the Fourier Transform applicable to finite-length, discrete-time signals. It samples the DTFT spectrum at discrete frequency points. For a signal $x[n]$ of length N , its DFT $X[k]$ is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn} \quad \text{for } k = 0, 1, \dots, N-1 \quad (1)$$

The corresponding frequencies are $f_k = k \cdot \frac{f_s}{N}$.

- **Inverse Discrete Fourier Transform (IDFT):** The IDFT reconstructs the time-domain signal from its DFT coefficients:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi}{N} kn} \quad \text{for } n = 0, 1, \dots, N-1 \quad (2)$$

- **Power Spectral Density (PSD):** The PSD describes how the power of a signal is distributed over frequency. For a DFT $X[k]$, the PSD can be estimated as $|X[k]|^2/N$.

6 Theory

6.1 Electroencephalogram (EEG) Signals

- **Origin:** Electrical activity generated by synchronized neuron populations in the cerebral cortex.
- **Measurement:** Non-invasive recording via scalp electrodes placed according to standardized systems (e.g., 10-20 system).
- **Characteristics:** EEG signals are complex, microvolt-range signals mixed from multiple brain regions and often contaminated by artifacts.
- **EEG Frequency Bands:**
 - Delta (δ): 0.5–4 Hz
 - Theta (θ): 4–8 Hz
 - Alpha (α): 8–13 Hz
 - Beta (β): 13–30 Hz
 - Gamma (γ): > 30 Hz

6.2 Discrete Fourier Transform (DFT)

- Transforms a discrete signal $x[n]$ of length N into frequency domain components $X[k]$:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \dots, N-1 \quad (3)$$

- Corresponding frequencies: $f_k = k \cdot \frac{f_s}{N}$.
- For real-valued signals, the spectrum is conjugate symmetric: $X[k] = X^*[N-k]$, so only the first half (up to Nyquist frequency) is analyzed.

6.3 Signal Reconstruction using Inverse DFT (IDFT)

- Converts frequency-domain data back to time domain:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \dots, N-1 \quad (4)$$

- Selective reconstruction: By zeroing DFT coefficients outside desired frequency bands and preserving conjugate symmetry, IDFT reconstructs a filtered version of the signal (e.g., alpha and beta bands).

7 Implementation

This section outlines the manual application of Discrete Fourier Transform (DFT) and Inverse DFT (IDFT) to a real EEG signal segment for frequency analysis and reconstruction.

7.1 Tools and Libraries

- **Python 3.12, NumPy, Matplotlib, MNE**

7.2 Data Source

- **Dataset:** PhysioNet EEG Motor Movement/Imagery Dataset <https://physionet.org/content/eegmmidb/1.0.0/S001/>

7.3 Procedure

1. **Signal Selection:** A 5-second EEG segment was extracted from channel 0 starting at 10s.
2. **Manual DFT:** A custom function `compute_dft()` was used to compute DFT using its standard formula:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi kn/N}$$

3. **Frequency Analysis:** The magnitude spectrum revealed strong Alpha (8–13 Hz) and Beta (13–30 Hz) activity.
4. **Band Extraction:** Each EEG band (Delta to Gamma) was isolated by zeroing out DFT bins outside the target frequency range using `extract_band()`.
5. **Manual IDFT:** Band-filtered frequency components were transformed back to time domain using:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \cdot e^{j2\pi kn/N}$$

6. **Reconstruction:** The sum of bandwise reconstructed signals closely approximated the original, showing Alpha and Beta bands' dominance.

8 Code Segments

The following Python code was used for the analysis.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 !pip install mne
5 import mne
6 #downloaded from https://physionet.org/content/eegmmidb/1.0.0/S001/#
   files-panel
7 edf_file = '/content/drive/MyDrive/Colab Notebooks/eeg.edf'
8 eeg_data = mne.io.read_raw_edf(edf_file, preload=True)
9 eeg_data.pick_types(eeg=True)
10
11 sampling_rate = eeg_data.info['sfreq']
12
13 start_time = 10
14 segment_length = 5
15 start_idx = int(start_time * sampling_rate)
16 end_idx = int((start_time + segment_length) * sampling_rate)
17
18 signal_raw = eeg_data.get_data(picks=0)[0]
19 segment_data = signal_raw[start_idx:end_idx]
20 time_axis = np.linspace(0, segment_length, len(segment_data))
21
22 plt.figure(figsize=(10, 3))
23 plt.plot(time_axis, segment_data)
24 plt.title("EEG Signal Segment (Time Domain)")
25 plt.xlabel("Time (seconds)")
26 plt.ylabel("Amplitude")
27 plt.grid(True)
28 plt.savefig('eeg_original_signal.png')
29 plt.show()
30
31 def compute_dft(input_signal):
32     N = len(input_signal)
33     output_freq = []
34     for k in range(N):
35         value = 0
36         for n in range(N):
37             angle = -2 * np.pi * k * n / N
38             value += input_signal[n] * np.exp(1j * angle)
39         output_freq.append(value)
40     return np.array(output_freq)
41
42 frequency_data = compute_dft(segment_data)
43 frequencies = np.fft.fftfreq(len(segment_data), 1 / sampling_rate)
44
45 plt.plot(frequencies[:len(frequencies)//2], np.abs(frequency_data[:len(
   frequency_data)//2]))

```

```

46 plt.title("Frequency Spectrum of EEG Segment")
47 plt.xlabel("Frequency (Hz)")
48 plt.ylabel("Magnitude")
49 plt.grid(True)
50 plt.savefig('eeg_frequency_spectrum.png')
51 plt.show()
52
53 # Define EEG frequency bands
54 eeg_bands = {
55     "Delta": (0.5, 4),
56     "Theta": (4, 8),
57     "Alpha": (8, 13),
58     "Beta": (13, 30),
59     "Gamma": (30, 100)
60 }
61
62 def extract_band(freq_data, freqs, band_range):
63     filtered = np.zeros_like(freq_data, dtype=complex)
64     for i, f in enumerate(freqs):
65         if band_range[0] <= abs(f) <= band_range[1]:
66             filtered[i] = freq_data[i]
67     return filtered
68
69 bandwise_freq_data = {}
70 for band_name, band_range in eeg_bands.items():
71     bandwise_freq_data[band_name] = extract_band(frequency_data,
72     frequencies, band_range)
73
74 def compute_idft(freq_data):
75     N = len(freq_data)
76     output_time = []
77     for n in range(N):
78         value = 0
79         for k in range(N):
80             angle = 2 * np.pi * k * n / N
81             value += freq_data[k] * np.exp(1j * angle)
82         output_time.append(value.real / N)
83     return np.array(output_time)
84
85 for band_name, freq_band in bandwise_freq_data.items():
86     time_signal = compute_idft(freq_band)
87     plt.figure(figsize=(10, 3))
88     plt.plot(time_axis, time_signal)
89     plt.title(f"{band_name} Band (Time Domain)")
90     plt.xlabel("Time (s)")
91     plt.ylabel("Amplitude")
92     plt.grid(True)

```

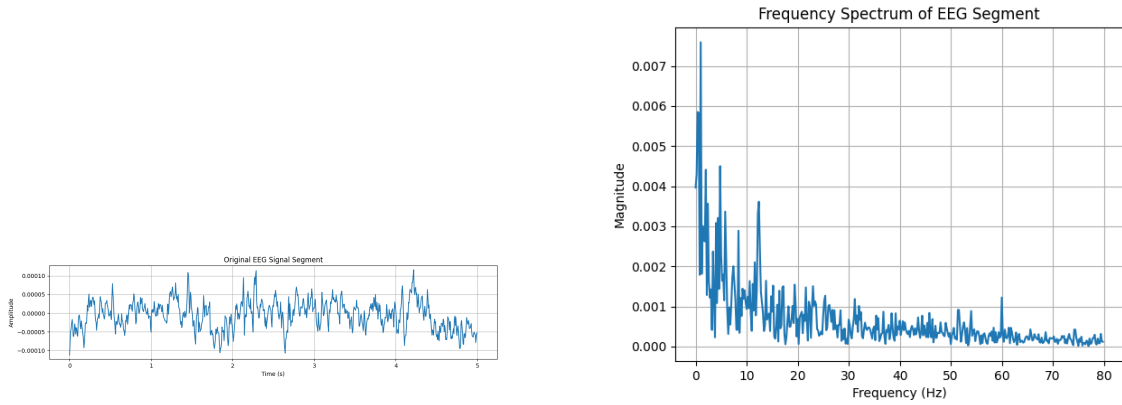
```

92     plt.show()
93
94     reconstructed_signal = sum(compute_idft(band) for band in
    bandwise_freq_data.values())
95 plt.figure(figsize=(10, 4))
96 plt.plot(time_axis, segment_data, label="Original", alpha=0.7)
97 plt.plot(time_axis, reconstructed_signal, label="Reconstructed",
    linestyle='--')
98 plt.title("Original vs. Reconstructed EEG Signal")
99 plt.xlabel("Time (s)")
100 plt.ylabel("Amplitude")
101 plt.legend()
102 plt.grid(True)
103 plt.savefig("Original vs. Reconstructed EEG Signal.png")
104 plt.show()

```

Listing 1: Python code for DFT analysis of EEG signal

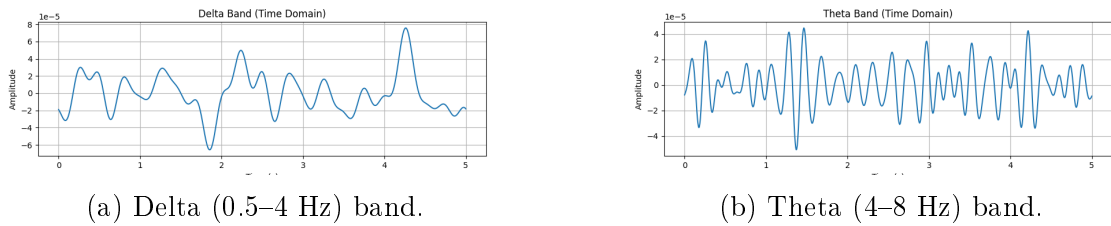
9 Output/Results



(a) EEG signal in time domain.

(b) Frequency spectrum of the EEG signal.

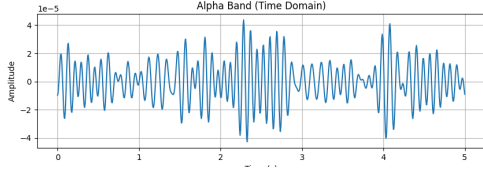
Figure 1: Time and frequency domain representations of the EEG signal.



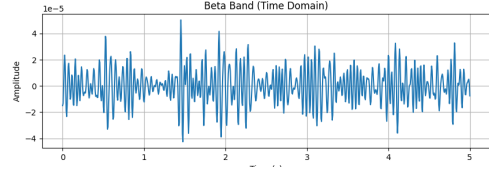
(a) Delta (0.5–4 Hz) band.

(b) Theta (4–8 Hz) band.

Figure 2: Reconstructed EEG signals from Delta and Theta bands.



(a) Alpha (8–13 Hz) band.



(b) Beta (13–30 Hz) band.

Figure 3: Reconstructed EEG signals from Alpha and Beta bands.

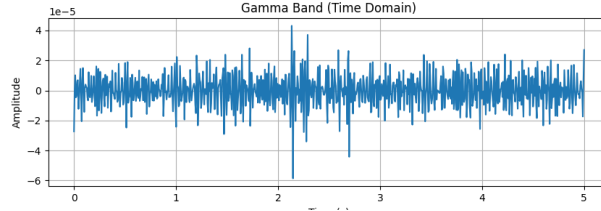


Figure 4: Reconstructed EEG signal from the Gamma (30–100 Hz) band.

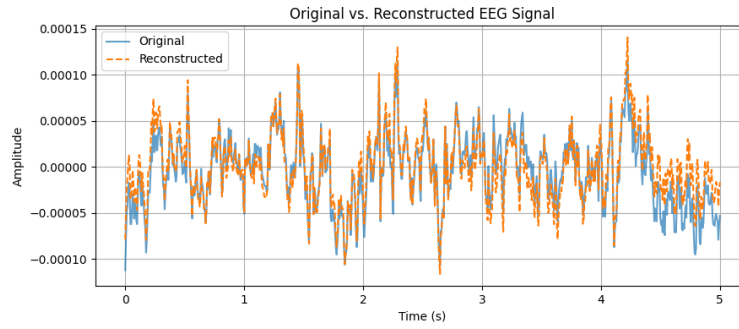


Figure 5: Comparison of the original EEG signal and the reconstructed signal.

10 Conclusion

This assignment demonstrated the manual application of the Discrete Fourier Transform (DFT) and Inverse DFT (IDFT) on a real EEG signal segment. Key frequency components were identified in the Alpha (8–13 Hz) and Beta (13–30 Hz) bands. The reconstructed signal, using only these bands, retained the core features of the original signal, highlighting the dominance of these frequencies.

This analysis shows that DFT is a useful tool for extracting meaningful brainwave patterns. However, the method was applied to a short segment and assumed signal stationarity, which may limit generalization. Future improvements may involve analyzing longer signals, multiple channels, and using time-frequency methods like STFT or wavelets.